DIAdem API Reference

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Calculator Functions

Calculator Functions

The following topics describe the functions of the DIAdem calculator.

Calculator Functions > Input and Result Quantities in Quantity-Sized Calculations

Input and Result Quantities in Quantity-Sized Calculations

To calculate quantity-based in DIAdem, click the **Calculate Quantity-Based** button on the DIAdem ANALYSIS toolbar. When executing quantity-based calculations, DIAdem checks whether the selected physical quantities can be used in the same calculation. If DIAdem cannot execute the calculation due to inappropriate physical quantities, DIAdem displays an error message. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Additional rules apply for the Calculator and the `Calculate` command. Refer to the Help page of the Calculate command for further information.

Some functions need dimensionless input quantities or return dimensionless result quantities. These physical quantities all have the exponent `0` for all base units. `1`, `%`, and `dB` are examples of dimensionless physical sizes. If you do not specify a result quantity, DIAdem uses the unit `1` as the result unit for these functions.

For most VBS functions the physical input quantity is arbitrary and the physical result quantity dimensionless. The following functions however need a special physical input quantity or return special physical result quantities:

Numeric Functions

+, -, Abs, Frac, MaxV, MinV, Random, Rnd,	The physical input quantities and the result quantities are the same.

Round, Trunc,	
*, -, ^, Sqr, Sqrt	DIAdem specifies the physical result quantity with the physical dimensions of the input quantities.
Exp, Eex, Fak, Lg, Ln, Log,	The input quantity must be dimensionless. The physical result quantity is also dimensionless.
Mod	The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same.
Sign, SND, Rtp, Rtt	The physical input quantity is arbitrary. The physical result quantity is dimensionless.

Trigonometric Functions

ArcCos, ArcCot, ArcSin, ArcTan,	The input quantity must be dimensionless. The result quantity has the physical quantity `Plane angle`.
ArcTanH, Cos	The input quantity must be dimensionless. The physical result quantity is also dimensionless.
ATanK	The first and second physical input quantity must be the same. The result quantity has the physical quantity `Plane angle`.
Deg	The input quantity must have the unit `rad`. The result quantity has the physical quantity `Plane angle`.
Rad	The input quantity must have the unit `deg`. The result quantity has the physical quantity `Plane angle`.
Cos, CosH, Cot, Sin, SinH, Tan, TanH	The input quantity must have the physical quantity `Plane angle`. The physical result quantity is dimensionless.

Boolean Functions

AND, NOT, OR, =, >, >=, <, <=, <>	The first and the second physical quantity must be the same. The physical result quantity is dimensionless.

Bit Functions

AndB, OrB, XorB	The first and the second physical quantity must be the same. The physical input quantities and the result quantities are the same.
ClrB, GetB, NotB, SetB, ShL, ShR,	The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same.

Calculator Functions > Numeric Functions

Numeric Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the numeric functions in DIAdem. You can use these functions, for example, to calculate the root, the square, and the logarithm of a number, or to determine random numbers.

Function: Abs

Function: Abs

Calculates the absolute value of a number.

```
ReturnValue = Abs(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns positive numeric values.

Examples

The following example calculates the absolute value of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Abs(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The physical input quantities and the result quantities are the same.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Deg

Function: Deg

Converts an angle from radians to degrees.

```
ReturnValue = Deg(Angle)
```

Input Parameters

Angle	Specifies a numeric value or a variable in radians, or a data channel that contains radians.

Return Parameters

ReturnValue	Returns angles in degrees.

Examples

The following example converts the value `Pi/3` to its angle value.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Deg(Pi/3)     'intMyResult = 60
```

The following example converts the values in the `Group1/Input` data channel from `radians` into `degrees` and saves the results in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Deg (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note Use the Rad function to convert angle values from degree to radians.

The input quantity must have the unit `rad`. The result quantity has the physical quantity `Plane angle`.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Calculator Functions > Numeric Functions > Function: Ceil

Function: Ceil

Specifies the integer part of a number, where the function rounds up the transferred value to obtain an integer number. For negative numbers the function returns integer numbers that are greater than or equal to the `Value` argument.

```
ReturnValue = Ceil(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns integer values.

Examples

The following example rounds up some values:

VBScript	Python

Copy script

```
Call MsgBox(Ceil(2.6))  '=> 3
Call MsgBox(Ceil(2.1))  '=> 3
Call MsgBox(Ceil(-2.1)) '=> -2
Call MsgBox(Ceil(-2.6)) '=> -2
```

The following example rounds up the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Ceil (Ch(""Group1/Input""))")
```

	Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The physical input quantities and the result quantities are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Topics

Floor | Int | Round | Trunc

Function: Eex

Function: Eex

Exponentiates a number to the base `10`.

```
ReturnValue = Eex(Exponent)
```

Input Parameters

Exponent	Specifies a numeric value or a variable as an exponent, or a data channel that contains exponents.

Return Parameters

ReturnValue	Returns positive values.

Examples

The following example calculates the third power of `10`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Eex(3)     'intMyResult = 1000
```

The following example exponentiates the values of the `Group1/Input` data channel to the base `10` and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Eex (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must be dimensionless. The physical result quantity is also dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd |

Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Exp

Function: Exp

Exponentiates a number to the base e (Euler number).

```
ReturnValue = Exp(Exponent)
```

Input Parameters

Exponent	Specifies a numeric value or a variable as an exponent, or a data channel that contains exponents.

Return Parameters

ReturnValue	Returns positive values.

Examples

The following example exponentiates the values of the `Group1/Input` data channel to the base e (Euler number) and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Exp (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note DIAdem only uses the DIAdem function `Exp` if you use the FormulaCalc command otherwise DIAdem uses the VBS function Exp.

The input quantity must be dimensionless. The physical result quantity is also dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Fak

Function: Fak

Calculates the factorial of a number.

```
ReturnValue = Fak(Value)
```

Input Parameters

Value	Specifies an integer value, a variable, or a data channel that contains integer values.

Return Parameters

ReturnValue	Returns integer values.

Examples

The following example calculates the factorial of the number `6`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Fak(6)     'intMyResult = 720
```

The following example calculates the factorial of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Fak (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must be dimensionless. The physical result quantity is also dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview

	of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Floor

Function: Floor

Specifies the integer part of a number, where the function rounds down the transferred value to obtain an integer number. For negative numbers the function returns integer numbers that are smaller than or equal to the `Value` argument.

```
ReturnValue = Floor(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns integer values.

Examples

The following example rounds down some values:

VBScript	Python

Copy script

```
Call MsgBox(Floor(2.6))  '=> 2
Call MsgBox(Floor(2.1))  '=> 2
Call MsgBox(Floor(-2.1)) '=> -3
Call MsgBox(Floor(-2.6)) '=> -3
```

The following example rounds down the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Floor (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The physical input quantities and the result quantities are the same.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Topics

Ceil | Int | Round | Trunc

Function: Frac

Function: Frac

Determines the decimal places of a number.

```
ReturnValue = Frac(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns the decimal places as numeric values. For invalid values the value is `NoValue`.

Examples

The following example determines the decimal places of the number `Pi`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Frac(Pi)     'intMyResult = 0.14159265358979
```

The following example determines the decimal places of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Frac (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note Use the Trunc function to determine the integer part of a number.

The physical input quantities and the result quantities are the same.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Lg

Function: Lg

Calculates the logarithm with a base of `10`. DIAdem uses the DIAdem function `Lg` only if you use the FormulaCalc command in your script, otherwise DIAdem uses the VBS function Log.

```
ReturnValue = Lg(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the decadic logarithm of the number `1000`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Lg(1000)     'intMyResult = 3
```

The following example calculates the decadic logarithm of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Lg (Ch(""Group1/Input""))")
```

	Note You must use the commands Calculate or FormulaCalc in scripts in order to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The input quantity must be dimensionless. The physical result quantity is also dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Ln

Function: Ln

Calculates the logarithm with a base of ∈ (Eularian number).

```
ReturnValue = Ln(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the natural logarithm of the number `Pi`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Ln(Pi)     'intMyResult = 1.1447298858494
```

The following example calculates the natural logarithm of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Ln (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must be dimensionless. The physical result quantity is also dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: MaxV

Function: MaxV

Determines the greater of two numbers.

```
ReturnValue = MaxV(Value1, Value2)
```

Input Parameters

Value1	Specifies a numeric value, a variable, or a data channel.
Value2	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns numeric values.

Note Use the command ValMax instead of the function `MaxV` if you are not calculating quantity-based because the `ValMax` command is faster.

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The physical input quantities and the result quantities are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Command: ValEqual | Command: ValEqualGT | Command: ValEqualLT | Command: ValGT | Command: ValLT | Command: ValMax | Command: ValMin | Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MinV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: MinV

Function: MinV

Determines the smaller of two numbers.

```
ReturnValue = MinV(Value1, Value2)
```

Input Parameters

Value1	Specifies a numeric value, a variable, or a data channel.
Value2	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns numeric values.

	Note Use the command ValMin instead of the function `MinV` if you are not calculating

	quantity-based because the `ValMin` command is faster.

	The physical input quantities and the result quantities are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Command: ValEqual | Command: ValEqualGT | Command: ValEqualLT | Command: ValGT | Command: ValLT | Command: ValMax | Command: ValMin | Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MaxV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Mod

Function: Mod

Divides two numeric values as integers and returns the remainder. DIAdem only uses the DIAdem function `Mod` if you use the FormulaCalc command in your script, otherwise DIAdem uses the VBS function Mod, which rounds the input parameters to integers, divides the numbers,and returns the divider rest.

ReturnValue = Dividend `Mod` ***Divisor***

Input Parameters

Dividend	Specifies a numeric value, a variable, or a data channel for the dividend.
Divisor	Specifies a numeric value, a variable, or a data channel for the divisor.

Return Parameters

Returns the remainder as a numeric value.

Examples

The following example divides the values in each line of the `Group1/Input` and `Group1/Reference` data channels, and saves the remainders in the `Group2/Result` data channel:

VBScript	Python

Copy script

```
Call FormulaCalc ("Ch('Group2/Result'):= Ch('Group1/Input') Mod Ch('Group1/
Reference')")
```

The following example shows the differences between the DIAdem `Mod` function and the VBS `Mod` function:

VBScript	Python

Copy script

```
Call FormulaCalc  ("R1 := (e)Mod(Pi)")   '=> R1 = 2.71828182845905
Call Calculate ("R1 =  (e)Mod(Pi)")   '=> R1 = 0
```

Note You must use the commands Calculate or FormulaCalc in scripts in order to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: ^

Function: ^

Calculates the power of one number to another.

```
ReturnValue = Base ^ Exponent
```

Input Parameters

Base	Specifies a numeric value, a variable, or a data channel as the base.
Exponent	Specifies a numeric value, a variable, or a data channel as the exponent.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example exponentiates the values of the `Group1/Input` data channel to the base `2` and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= 2 ^ Ch(""Group1/Input"")")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note If you use the ^ operator in your script without the `Calculate` command, DIAdem applies the VBS exponential operator ^.

Related Functions

Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Rad

Function: Rad

Converts an angle from degrees to radians.

```
ReturnValue = Rad(Angle)
```

Input Parameters

Angle	Specifies a numeric value or a variable as an angle value, or a data channel that contains angle values.

Return Parameters

ReturnValue	Returns angle values with the unit radians.

Examples

The following example converts the angle value `45°` into radians.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Rad(45)    'intMyResult = 0.785398163397448
```

The following example converts the angle values of the `Group1/Input` data channel to radians and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Rad (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note Use the Deg function to convert angle values from radians to degrees.

The input quantity must have the unit `deg`. The result quantity has the physical quantity `Plane angle`.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Random

Function: Random

Determines random numbers in the value range from zero up to a specified value.

```
ReturnValue = Random(Limit)
```

Input Parameters

Limit	Specifies a numeric value, a variable, or a data channel, to limit the value range.

Return Parameters

ReturnValue	Returns random numeric values.

Examples

The following example determines a random number between `0` and `5`.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = Random(5)
```

The following example determines for each value of the `Group1/Input` data channel a random value between `0` and the channel value, and saves the random numbers in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Random (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The physical input quantities and the result quantities are the same.

	The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Rnd

Function: Rnd

Determines random numbers in the value range from zero up to a specified value. DIAdem uses the DIAdem function `Rnd` only if you use the FormulaCalc command in your script otherwise DIAdem uses the VBS function Rnd. Use the Random function in scripts.

```
ReturnValue = Rnd([Limit])
```

Input Parameters

Limit	Specifies a numeric value, a variable, or a data channel, to limit the value range.

Return Parameters

ReturnValue	Returns random numeric values.

Examples

The following example determines for each value of the `Group1/Input` data channel a random value between `0` and the channel value, and saves the random numbers in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call FormulaCalc ("Ch('Group2/Result'):= Rnd(Ch('Group1/Input'))")
```

Note You must use the commands Calculate or FormulaCalc in scripts in order to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The physical input quantities and the result quantities are the same.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Round

Function: Round

Rounds off a number to the nearest integer. DIAdem uses the DIAdem function `Round` only if you use the FormulaCalc command in your script otherwise DIAdem uses the VBS function Round.

```
ReturnValue = Round(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns integer values.

Examples

The following example rounds off the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call FormulaCalc ("Ch('Group2/Result'):= Round ('Group1/Input')")
```

Note You must use the commands Calculate or FormulaCalc in scripts in order to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The physical input quantities and the result quantities are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: RTP (Real To Part of Date)

Function: RTP (Real To Part of Date)

Converts a number in DIAdem double time format into a text in time format and determines from the calculated date partial date information based on the variable ApplicationTimebaseHighResolution.

```
ReturnValue = RTP(Value, Text)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel containing the elapsed seconds. You can also specify a variant value with the subtype `Date` such as the functions CDate or Now return. **Note** If you save time values in DIAdem that are VBS VBDate subtype variables, the high resolution is lost. VBS only offers time values for this data type which provides only a limited resolution.
Text	Specifies the wildcard of the time format for the date information you want to extract, such as the day or year.

Return Parameters

ReturnValue	Returns the date information you want. The `RTP` function returns the result `0` for negative values.

Example

The following example converts the values of the `Group1/Input` data channel into time format, determines the months, and saves this result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= RTP (Ch(""Group1/Input""), ""M"")")
```

	Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The physical input quantity is arbitrary. The physical result quantity is dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTT (Real to Text) | Function: Sign |

Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: RTT (Real to Text)

Function: RTT (Real to Text)

Converts a number that exists in the DIAdem double time format into a text. The interpretation of the number depends on the variable ApplicationTimebaseHighResolution.

```
ReturnValue = RTT(Value[, Format])
```

Input Parameters

Value	Specifies a numeric value or a variable in DIAdem double time format. **Note** If you specify Variant values with the `Date` subtype, such as those returned by the CDate or Now functions, these values are automatically converted to DIAdem time format and lose their high resolution.
[***Format***]	Specifies a format definition for the time format.

Return Parameters

ReturnValue	Returns the date as text.

The TTR function converts a text in time format into a number in DIAdem double time format.

Examples

The following example converts a positive number into time format.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = RTT(14*3600)
```

The following example formats the current date.

VBScript	Python

Copy script

```
Call MsgBox(RTT(Now,"#YYYY/MM/DD"))
```

	The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Command: ChnConvertNumericToTime | Command: ChnConvertTimeToNumeric | Command: ConvertTimeType | Command: TTD | Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTP (Real To Part of Date) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: TTR (Text to Real) | Function: VU

Function: Sign

Function: Sign

Specifies the sign of a number.

```
ReturnValue = Sign(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Receives the sign of `number`. The value is `-1` for negative values, `0` for the value `0`, and `1` for positive values. For invalid values the value is `NoValue`.

Examples

The following example determines the sign for the value `-Pi`.

VBScript	Python

Copy script

```
Dim lngMyResult
lngMyResult = Sign(-Pi)     'lngMyResult = -1
```

The following example determines the signs of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Sign(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The physical input quantity is arbitrary. The physical result quantity is dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: SND

Function: SND

Calculates the function value of the standard normal distribution for a number.

```
ReturnValue = SND(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns function values of the standard normal distribution.

Examples

The following example determines the function value of the standard normal distribution for the value `0`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = SND(0)     'intMyResult = 0.5
```

The following example determines the function values of the standard normal distribution for the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= SND (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The physical input quantity is arbitrary. The physical result quantity is dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: Sqr | Function: Sqrt | Function: Trunc | Function: VU

Function: Sqr

Function: Sqr

Calculates the square of a number. DIAdem uses the DIAdem function `Sqr` if you use the FormulaCalc command in your script, otherwise DIAdem uses the VBS function Sqr, which calculates the square root.

```
ReturnValue = SQR(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns positive values.

The following example squares the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call FormulaCalc ("Ch('Group2/Result'):= SQR ('Group1/Input')")
```

Note You must use the commands Calculate or FormulaCalc in scripts in order to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

DIAdem specifies the physical result quantity with the physical dimensions of the input quantities.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqrt | Function: Trunc | Function: VU

Function: Sqrt

Function: Sqrt

Calculates the square root of a number.

```
ReturnValue = Sqrt(Value)
```

Input Parameters

Value	Specifies a positive numeric value, a variable that contains a positive value, or a data channel that contains positive values.

Return Parameters

ReturnValue	Returns positive numeric values.

Examples

The following example calculates the square root of a number.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Sqrt(6.5109)    'intMyResult = 2.55164652724471
```

The following example calculates the square root of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Sqrt (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the

	procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	Note You also can use the VBS function Sqr to calculate the square root of a value or of a variable in a script.

	DIAdem specifies the physical result quantity with the physical dimensions of the input quantities. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Trunc | Function: VU

Function: Trunc

Function: Trunc

Determines the integer part of a number.

```
ReturnValue = Trunc(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

Return Parameters

ReturnValue	Returns integer values. For invalid values the value is `NoValue`.

Examples

The following example determines the integer part of a real number.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Trunc(-75.5109)     'intMyResult = -75
```

The following example determines the integer parts of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Trunc (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note Use the Frac function to determine the decimal places of a number.

	The physical input quantities and the result quantities are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: VU

Function: VU

Function: VU

Connects a value and a unit.

```
ReturnValue = VU(Value, [Unit])
```

Input Parameters

Value	Specifies a numeric value or a variable.
[Unit]	Specifies a unit. If you do not specify the unit, DIAdem uses the unit "1".

Return Parameters

DIAdem saves the result unit in a quantity-based calculation of the Calculate command in the CalculateResultUnit variable. Refer to the Calculate command for rules for input units and result units.

Examples

The following example calculates the quotient from "10 m/s" and "2 ms" and displays the result value and the result unit as a message.

VBScript	Python

Copy script

```
CalcQuantityBased = TRUE
Call Calculate("R1=VU(10, ""m/s"") / VU(2, ""ms"")")
Call MsgBoxDisp("Result: Value " & R1 & " with unit " & CalculateResultUnit)
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Related Functions

Function: ^ | Function: Abs | Function: Ceil | Function: Deg | Function: Eex | Function: Exp | Function: Fak | Function: Floor | Function: Frac | Function: Lg | Function: Ln | Function: MaxV | Function: MinV | Function: Mod | Function: Rad | Function: Random | Function: Rnd | Function: Round | Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Sign | Function: SND | Function: Sqr | Function: Sqrt | Function: Trunc

Calculator Functions > Trigonometric Functions

Trigonometric Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the trigonometric DIAdem functions. You can use these functions,for example, to calculate the sine or the cosine of a number.

Function: ArcCos

Function: ArcCos

Valid names: ArcCos, ACos

Calculates the arc cosine of a number.

```
ReturnValue = ArcCos(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the cosine values. Value range from -1 to +1.

Return Parameters

ReturnValue	Returns an angle in radians from 0 to Pi.

Examples

The following example calculates from the arc cosine of a number the corresponding angle value in degrees.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = Deg(ACos(-0.9143))    'dblMyResult = 156.10651
```

The following example calculates the arc cosine of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ACos(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must be dimensionless. The result quantity has the physical quantity `Plane angle`.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: ArcCot

Function: ArcCot

Valid names: ArcCot, ACot

Calculates the arccotangent of a number.

```
ReturnValue = ArcTan(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the cotangent values.

Return Parameters

ReturnValue	Returns an angle in radians from -Pi/2 to Pi/2.

Examples

The following example calculates the arccotangent of a number.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = ACot(1.9143)    'dblMyResult = 0.481394161422127
```

The following example calculates the arc tangent of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ACot(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the

	procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The input quantity must be dimensionless. The result quantity has the physical quantity `Plane angle`. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: ArcSin

Function: ArcSin

Valid names: ArcSin, ASin

Calculates the arc sine of a number.

```
ReturnValue = ArcSin(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the sine values. Value range from -1 to +1.

Return Parameters

ReturnValue	Returns an angle in radians from -Pi/2 to Pi/2.

Examples

The following example calculates the corresponding angle value in degrees from the arc sine of a number.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = Deg(ASin(0.9143))    'dblMyResult = 66.1065125606145
```

The following example calculates the arc sine of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ASin(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must be dimensionless. The result quantity has the physical quantity `Plane angle`.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: ArcTan

Function: ArcTan

Valid names: ArcTan, ATan

Calculates the arc tangent of a number.

```
ReturnValue = ArcTan(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the tangent values.

Return Parameters

ReturnValue	Returns an angle in radians from -Pi/2 to Pi/2.

Examples

The following example calculates the arc tangent of a number.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = ATan(1.9143)     'dblMyResult = 1.0894
```

The following example calculates the arc tangent of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ATan(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note You also can use the VBS function Atn to calculate the arc tangent of a number.

The input quantity must be dimensionless. The result quantity has the physical quantity `Plane angle`.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: ArcTanH

Function: ArcTanH

Valid names: ArcTanH, ATanH

Calculates the hyperbolic arcustangens of a number.

```
ReturnValue = ArcTanH(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel. Value range from -1 to +1.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the hyperbolic arc tangent of a number.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = ATanH(0.9143)    'dblMyResult = 1.5531
```

The following example calculates the hyperbolic arc tangent of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ATanH(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must be dimensionless. The physical result quantity is also dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: ATan2

Function: ATan2

Calculates the arc tangent from two numbers. Specify the adjacent leg and the opposite leg to assign the result to any of the four quadrants and handle 0°, 90°, and 270°, which are special cases.

```
ReturnValue = ATan2(Opposite, Adjacent)
```

Input Parameters

Opposite	Specifies a numeric value, a variable, or a data channel that contains the opposite leg.

Adjacent	Specifies a numeric value, a variable, or a data channel that contains the adjacent leg.

Return Parameters

ReturnValue	Returns an angle in radians from -Pi to Pi.

Examples

The following example calculates the arc tangent of two numbers.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = ATan2(12.9143,17.209)    'dblMyResult = 0.64378
```

The following example calculates the arc tangent of the values in the data channels `Group1/Input` and `Group1/Input2` and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ATan2(Ch(""Group1/Input""),Ch(""Group1/
Input2""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the

	procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The first and second physical input quantity must be the same. The result quantity has the physical quantity `Plane angle`. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: ATanK

Function: ATanK

Possible spellings: ATanK, ATK

Calculates the arc tangent from two numbers. Specify the adjacent leg and the opposite leg to assign the result to any of the four quadrants and handle 0°, 90°, and 270°, which are special cases.

```
ReturnValue = ATanK(Opposite, Adjacent)
```

Input Parameters

Opposite	Specifies a numeric value, a variable, or a data channel that contains the opposite leg.
Adjacent	Specifies a numeric value, a variable, or a data channel that contains the adjacent leg.

Return Parameters

ReturnValue	Returns an angle in radians from 0 to 2*Pi.

Examples

The following example calculates the arc tangent of two numbers.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = ATanK(12.9143,17.209)    'dblMyResult = 0.64378
```

The following example calculates the arc tangent of the values in the data channels `Group1/Input` and `Group1/Input2` and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ATanK(Ch(""Group1/Input""),Ch(""Group1/
Input2""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

	The first and second physical input quantity must be the same. The result quantity has the physical quantity `Plane angle`. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: Cos

Function: Cos

Calculates the cosine of a number.

```
ReturnValue = Cos(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the angles in radians.

Return Parameters

ReturnValue	Returns numeric values ranging from -1 to +1.

Examples

The following example calculates the cosine of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Cos(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must be dimensionless. The physical result quantity is also dimensionless. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: CosH

Function: CosH

Calculates the hyperbolic cosine of a number.

```
ReturnValue = CosH(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the angles in radians.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the hyperbolic cosine of a number.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = CosH(1.9143)     'dblMyResult = 3.46482
```

The following example calculates the hyperbolic cosine of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= CosH(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must have the physical quantity `Plane angle`. The physical result quantity is dimensionless.

	The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: Cot | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: Cot

Function: Cot

Calculates the cotangent of a number.

```
ReturnValue = Cot(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the angles in radians.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the cotangent of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Cot(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must have the physical quantity `Plane angle`. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Sin | Function: SinH | Function: Tan | Function: TanH

Function: Sin

Function: Sin

Calculates the sine of a number.

```
ReturnValue = Sin(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the angles in radians.

Return Parameters

ReturnValue	Returns numeric values ranging from -1 to +1.

Examples

The following example calculates the sine of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Sin(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note DIAdem only uses the DIAdem function `Sin` if you use the FormulaCalc command otherwise DIAdem uses the VBS function Sin.

The input quantity must have the physical quantity `Plane angle`. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function:

Cot | Function: SinH | Function: Tan | Function: TanH

Function: SinH

Function: SinH

Calculates the hyperbolic sine of a number.

```
ReturnValue = SinH(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the angles in radians.

Output Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the hyperbolic sine of a number.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = SinH(1.9143)    'dblMyResult = 3.31737229713868
```

The following example calculates the hyperbolic sine of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= SinH(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must have the physical quantity `Plane angle`. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: Tan | Function: TanH

Function: Tan

Function: Tan

Calculates the tangent of a number.

```
ReturnValue = Tan(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the angles in radians.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the tangent of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= Tan(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note DIAdem only uses the DIAdem function `Tan` if you use the FormulaCalc command in your script otherwise DIAdem uses the VBS function Tan.

The input quantity must have the physical quantity `Plane angle`. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: TanH

Function: TanH

Function: TanH

Calculates the hyperbolic tangent of a number.

```
ReturnValue = TanH(Value)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel that contains the angles in radians.

Return Parameters

ReturnValue	Returns numeric values.

Examples

The following example calculates the hyperbolic tangent of a number.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = TanH(1.9143)     'intMyResult = 0.957445
```

The following example calculates the hyperbolic tangent of the values in the `Group1/Input` data channel and saves the result in the `Group2/Result` data channel.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= TanH(Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The input quantity must have the physical quantity `Plane angle`. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ArcCos | Function: ArcCot | Function: ArcSin | Function: ArcTan | Function: ArcTanH | Function: ATan2 | Function: ATanK | Function: Cos | Function: CosH | Function: Cot | Function: Sin | Function: SinH | Function: Tan

Calculator Functions > Boolean Functions

Boolean Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the Boolean DIAdem functions. You can use these functions to negate values and for logical AND or OR operations.

Function: And

Function: And

Calculates the And operation of two values.

```
ReturnValue = Value1 And Value2
```

Parameters

Value1	Specifies a Boolean value, a Boolean variable, or a data channel.
Value2	Specifies a Boolean value, a Boolean variable, or a data channel.

Result

ReturnValue	Returns Boolean values.

Value1		Value2	Result
TRUE	and	TRUE	TRUE
TRUE	and	1	TRUE
23	and	17	TRUE
FALSE	and	TRUE	FALSE
FALSE	and	FALSE	FALSE
FALSE	and	0	FALSE
FALSE	and	-100	FALSE
0	and	0	0
0	and	TRUE	0
0	and	23	0

Examples

The following example checks rowwise whether the two data channels contain values. If both data channels have a value in the same row, the result in the new channel `Group2/Result` is the value `1`, and if either of the two data channels has the value `0` in the same row, the result is the value `0`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= (Ch(""Group1/Input""))And(Ch(""Group1/Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note If you use the `And` function without the `Calculate` command in your script, DIAdem uses the VBS operator And, which compares two integer values bitwise.

The first and the second physical quantity must be the same. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: CTNV | Function: IIf | Function: NOT | Function: OR | Function: VIT

Function: NOT

Function: NOT

Computes the logical negation.

```
ReturnValue = NOT Value
```

Input Parameters

Value	Specifies a Boolean value, a Boolean variable, or a data channel.

Return Parameters

ReturnValue	Returns Boolean values. TRUE — For the value FALSE FALSE — For the value TRUE 1 — For the value zero

Examples

The following example checks the `Group1/Input` data channel for values. The result in the `Group2/Result` data channel returns the value `0` for every value, and returns the value `0` for the value `1`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= NOT (Ch(""Group1/Input""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note DIAdem only uses the DIAdem function `Not` if you use the FormulaCalc command otherwise DIAdem uses the VBS function Not.

The first and the second physical quantity must be the same. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: And | Function: CTNV | Function: IIf | Function: OR | Function: VIT

Function: OR

Function: OR

Executes the OR operation on two values.

```
ReturnValue = Value1 OR Value2
```

Input Parameters

Value1	Specifies a Boolean value, a Boolean variable, or a data channel.
Value2	Specifies a Boolean value, a Boolean variable, or a data channel.

Return Parameters

ReturnValue	Returns Boolean values.

Value1		Value2	Result
TRUE	or	TRUE	TRUE
TRUE	or	FALSE	TRUE
FALSE	or	1	TRUE
FALSE	or	17	TRUE
TRUE	or	-100	TRUE
TRUE	or	0	TRUE
0	or	1	TRUE
FALSE	or	FALSE	FALSE
0	or	0	0
0	or	FALSE	0

Examples

The following example checks rowwise whether the two data channels contain values. If one of the two data channels has a value in the same row, the result in the new channel `Group2/Result` returns the value `1`, and if both data channels have the value `0` in the same row, the result returns the value `0`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= (Ch(""Group1/Input"")) OR (Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note If you use the `Or` function without the `Calculate` command in your script, DIAdem uses the VBS operator Or, which computes two integer values bitwise.

The first and the second physical quantity must be the same. The physical result quantity is dimensionless.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: And | Function: CTNV | Function: IIf | Function: NOT | Function: VIT

Calculator Functions > Bit Functions

Bit Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the DIAdem bit functions. You can use these functions to set, check, negate, or shift bits, for example.

Function: AndB

Function: AndB

Calculates the number that results from the identical bits of two numbers. The `AndB` function converts the decimal numbers into binary numbers, determines the identical bits, and reconverts this binary number into a decimal number.

```
ReturnValue = AndB(Value1, Value2)
```

Input Parameters

Value1	Specifies a numeric value, a variable, or a data channel.
Value2	Specifies a numeric value, a variable, or a data channel.

Note If you specify numbers with decimal places for the parameters `Value1` and `Value2`, DIAdem rounds the numbers to integers.

Return Parameters

ReturnValue	Returns integer values. If two numbers contain no identical bits, the function returns the result `0`. If the input value is `NoValue`, the `AndB` function returns the value `NoValue`.

Examples

The following example calculates the identical bits of the two decimal numbers `10` and `12` and saves the corresponding decimal number.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = AndB(10,12)    'intMyResult = 8
```

The following example calculates the identical bits of the values in each line of two data channels, and saves the decimal numbers in the new data channel `Group2/Result`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= AndB (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note You also can use VBS function And to calculate the identical bits of two numbers.

The first and the second physical quantity must be the same. The physical input quantities and the result quantities are the same.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: ClrB | Function: GetB | Function: NotB | Function: OrB | Function: SetB | Function: ShL | Function: ShR | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: ClrB

Function: ClrB

Deletes a specific bit from the binary display of a decimal number. The `ClrB` function converts a decimal number to a binary number, deletes the specified bit, and reconverts the new binary number to a decimal number.

```
ReturnValue = ClrB(Value, BitIndex)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.
BitIndex	Specifies a numeric value, a variable, or a data channel for the index of the bit you want to delete. Value range `0` to `31`

Note If you enter numbers with decimal places for the values ***Value*** and ***Index***, DIAdem rounds the numbers to integers.

Return Parameters

ReturnValue	Returns integer values. If the specified bit already has the value `0`, its value remains `0`. If the input value is `NoValue`, the `ClrB` function returns the value `NoValue`.

Examples

The following example deletes the second bit of the binary display of the decimal number `10`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = ClrB(10,1)     'intMyResult = 8
```

The following example deletes the bits indexed by the `Group1/Reference` data channel from the values in the `Group1/Input` data channel, and saves the new decimal numbers in the new data channel `Group2/Result`. DIAdem skips values of the `Group1/Reference` index channel that are higher than `31`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ClrB (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: GetB | Function: NotB | Function: OrB | Function: SetB | Function: ShL | Function: ShR | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: GetB

Function: GetB

Checks a bit you specify in the binary representation of a decimal number. The `GetB` function converts the decimal number into a dual number.

```
ReturnValue = GetB(Value, BitIndex)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.
BitIndex	Specifies a numeric value, a variable, or a data channel for the index of the bit you want to check. Value range 0 to 31

Note If you enter numbers with decimal places for the parameters `Value` and `BitIndex`, DIAdem rounds the numbers to integers.
From DIAdem 2014, the function no longer calculates the sum of the `BitIndex` input parameter.

Return Parameters

ReturnValue	Returns Boolean values. If the input value is `NoValue`, the `GetB` function returns the value `NoValue`. If the input parameter `BitIndex` is outside the value range, the function returns the value `0`.

Examples

The following example checks the second bit of the binary representation of the decimal number `10`.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = GetB(10,1)     'blnMyResult = TRUE
```

The following example checks the bits from the values in the `Group1/Input` data channel, indexed by the `Group1/Reference` data channel, and saves the result in the new data channel `Group2/Result`. DIAdem skips values of the `Group1/Reference` index channel that are higher than `31`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= GetB (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: ClrB | Function: NotB | Function: OrB | Function: SetB | Function: ShL | Function: ShR | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: NotB

Function: NotB

Negates the bits of the one-byte long binary representation of a decimal number. The `NotB` function converts the decimal numbers to one-byte long binary numbers, negates each bit, and reconverts the new binary number to a decimal number.

```
ReturnValue = NotB(Value, ByteLength)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.
ByteLength	Specifies a numeric value, a variable, or a data channel for the byte length. Value range 1 to 4

Note If you enter numbers with decimal places for the parameters `Value` and `ByteLength`, DIAdem rounds the numbers to integers.
From DIAdem 2014, the function no longer calculates the sum of the `ByteLength` input parameter.

Return Parameters

ReturnValue	Returns integer values. If the input value is `NoValue`, the `NotB` function returns the value `NoValue`. If the input parameter `ByteLength` is outside the value range, the function returns the value `0`.

Examples

The following example negates the one-byte long binary representation of the decimal number `10`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = NotB(10,1)     'intMyResult = 245
```

The following example switches the bits of the values in the `Group1/Input` data channel within the byte length given by the `Group1/Reference` data channel and saves the result in the new data channel `Group2/Result`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= NotB (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: ClrB | Function: GetB | Function: OrB | Function: SetB | Function: ShL | Function: ShR | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: OrB

Function: OrB

Calculates the number resulting from the shared bits of two numbers. The `OrB` function converts the decimal numbers into binary numbers, determines the shared bits, and reconverts this binary number into a decimal number.

```
ReturnValue = OrB(Value1, Value2)
```

Input Parameters

Value1	Specifies a numeric value, a variable, or a data channel.
Value2	Specifies a numeric value, a variable, or a data channel.

Note If you specify numbers with decimal places for the parameters `Value1` and `Value2`, DIAdem rounds the numbers to integers.

Return Parameters

ReturnValue	Returns integer values. If the input value is `NoValue`, the `OrB` function returns the value `NoValue`.

Examples

The following example calculates the shared bits of the two decimal numbers `12` and `16` and saves the corresponding decimal number.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = OrB(16,12)    'intMyResult = 28
```

The following example calculates the shared bits of the values of two data channels rowwise and saves the decimal numbers in the data channel `Group2/Result`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= OrB (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note You can also calculate the shared bits of two numbers with the VBS function Or.

The first and the second physical quantity must be the same. The physical input quantities and the result quantities are the same.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: ClrB | Function: GetB | Function: NotB | Function: SetB | Function: ShL | Function: ShR | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: SetB

Function: SetB

Sets a specific bit in the binary representation of a decimal number. The `SetB` function converts a decimal number to a binary number, sets the specified bit, and reconverts the new binary number to a decimal number.

```
ReturnValue = SetB(Value, BitIndex)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.

BitIndex	Specifies a numeric value, a variable, or a data channel for the index of the bit you want to check. Value range 0 to 31

	Note If you enter numbers with decimal places for the parameters `Value` and `BitIndex`, DIAdem rounds the numbers to integers. From DIAdem 2014, the function no longer calculates the sum of the `BitIndex` input parameter.

Return Parameters

ReturnValue	Returns integer values. If the indexed bit is already set, which means it has the value `1`, its value remains `1`. If the input value is `NoValue`, the `SetB` function returns the value `NoValue`. If the input parameter `BitIndex` is outside the value range, the function returns the value of the input parameter `Value`.

Examples

The following example sets the seventh bit in the binary representation of the decimal number `10`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = SetB(10,6)     'intMyResult = 74
```

The following example sets the bits, indexed by the `Group1/Reference` data channel, of the values in the `Group1/Input` data channel, and saves the new decimal numbers in the new data channel `Group2/Result`. DIAdem skips values of the `Group1/Reference` index channel that are higher than `31`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= SetB (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: ClrB | Function: GetB | Function: NotB | Function: OrB | Function: ShL | Function: ShR | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: ShL

Function: ShL

Shifts the bits in the binary representation of a decimal number to the left by a specific number of places. The `ShL` function converts the decimal number into a binary number, inserts on the right as many zeros as you specify, and reconverts the binary number into a decimal number.

```
ReturnValue = ShL(Value, Digits)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.
Digits	Specifies a numeric value, a variable, or a data channel for the number of places you want to shift. Value range 0 to 31

	Note If you enter numbers with decimal places for the parameters `Value` and `Digits`, DIAdem rounds off the numbers to integers. From DIAdem 2014, the function no longer calculates the sum of the `Digits` input parameter.

Return Parameters

ReturnValue	Returns integer values. If the input value is `NoValue`, the `ShLB` function returns the value `NoValue`. If the input parameter Digits is outside the value range, the function returns the value `0`.

Examples

The following example adds two zeros to the binary representation of the decimal number `13` and saves the new decimal number.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = ShL(13,2)     'intMyResult = 52
```

The following example adds bits on the right of the binary representation of the values

in the `Group1/Input` data channel. The number of places added is specified in the `Group1/Reference` data channel. The example saves the new decimal numbers in the data channel `Group2/Result`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ShL (Ch(""Group1/Input""),Ch(""Group1/Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: ClrB | Function: GetB | Function: NotB | Function: OrB | Function: SetB | Function: ShR | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: ShR

Function: ShR

Shifts the bits in the binary representation of a decimal number a specific number of places to the right. The `ShR` function converts the decimal number into a binary

number, deletes the specified number of places on the right of the binary number, and reconverts the new binary number into a decimal number.

```
ReturnValue = ShR(Value, Digits)
```

Input Parameters

Value	Specifies a numeric value, a variable, or a data channel.
Digits	Specifies a numeric value, a variable, or a data channel for the number of places you want to delete. Value range 0 to 31

	Note If you enter numbers with decimal places for the parameters `Value` and `BitIndex`, DIAdem rounds the numbers to integers. From DIAdem 2014, the function no longer calculates the sum of the `Digits` input parameter.

Return Parameters

ReturnValue	Returns integer values. If the input value is `NoValue`, the `ShRB` function returns the value `NoValue`. If the input parameter Digits is outside the value range, the function returns the value `0`.

Examples

The following example deletes the last two bits of the binary representation of the decimal number `13` and saves the new decimal number.

VBScript	Python

Copy script

```
Dim intMyResult
```

```
intMyResult = ShR(13,2)   'intMyResult = 3
```

The following example deletes bits on the right of the binary representation of the values in the `Group1/Input` data channel. The number of places deleted is specified in the `Group1/Reference` data channel. The example saves the new decimal numbers in the data channel `Group2/Result`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= ShR (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

The first physical input quantity is arbitrary. The second input quantity must be dimensionless. The first physical input quantity and the result quantity are the same. The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: ClrB | Function: GetB | Function: NotB | Function: OrB | Function: SetB | Function: ShL | Function: XOrB | Variable: ApplicationLegacyBinaryOperations

Function: XOrB

Function: XOrB

Determines the number resulting from the unequal bits of two numbers. The `XOrB` function converts the decimal numbers into binary numbers, determines the bits that the two numbers do not have in common, and reconverts this binary number into a decimal number.

```
ReturnValue = XOrB(Value1, Value2)
```

Input Parameters

Value1	Specifies a numeric value, a variable, or a data channel.
Value2	Specifies a numeric value, a variable, or a data channel.

Note If you specify numbers with decimal places for the parameters `Value1` and `Value2`, DIAdem rounds the numbers to integers.

Return Parameters

ReturnValue	Returns integer values. If the input value is `NoValue`, the `XOrB` function returns the value `NoValue`.

Examples

The following example determines the unequal bits of the decimal numbers `10` (dual: `1010`) and `12` (dual: `1100`).

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = XOrB(10,12)    'intMyResult = 6 (dual:0110)
```

The following example determines the unequal bits of the values in each line of two data channels, and saves the decimal numbers in the new data channel `Group2/Result`.

VBScript	Python

Copy script

```
Call Calculate ("Ch(""Group2/Result"")= XOrB (Ch(""Group1/Input""),Ch(""Group1/
Reference""))")
```

Note You must use the Calculate command in scripts to calculate data channels with formulas. Compute the data channels according to the syntax in the Calculator. Refer to the procedures Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Note You can also determine the unequal bits of two numbers with the VBS function XOr.

The first and the second physical quantity must be the same. The physical input quantities and the result quantities are the same.
The Input and Result Quantities for Quantity-Based Calculations page contains an overview of the input quantities and result quantities of all functions. Refer to the Help page Calculating Quantity-Based and Non-Quantity-Based in DIAdem for the conditions and rules for quantity-based calculations.

Related Functions

Function: AndB | Function: ClrB | Function: GetB | Function: NotB | Function: OrB | Function: SetB | Function: ShL | Function: ShR | Variable: ApplicationLegacyBinaryOperations

Text Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the DIAdem text functions. You can use these functions, for example, to determine the index of a text or the position of a character in a text, or to delete characters from a text.

Function: Char

Function: Char

Determines the character that corresponds to the entered ASCII code.

```
ReturnValue = Char(ASCIIcode)
```

Input Parameters

ASCIIcode	Specifies an integer value. Value range from 1 to 255.

Return Parameters

ReturnValue	Returns a text character.

Examples

The following example determines the character that corresponds to the ASCII code `65`.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = Char(65)     'strMyResult = A
```

Related Functions

Command: TTD | Function: FCNo | Function: FileNameSplit | Function: Idx | Function: Len | Function: PU | Function: Str | Function: Textaddition (+) | Function: TTR (Text to Real) | Function: Val

Function: FileNameSplit

Function: FileNameSplit

Valid names: NameSplit

Extracts information from a text.

```
ReturnValue = NameSplit(String, Type)
```

Input Parameters

String	Specifies a text or a text variable that contains a filename with extension and path.
Type	Specifies the wildcard for the file information you want to extract:

Value	Meaning
P	Path
N	Filename
E	Filename extension

Return Parameters

ReturnValue	Returns the filename, the path, or the extension as text. The function returns an empty text if the specified text does not contain the information you want.

Examples

The following examples extract various information from a file.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = NameSplit("C:\Folder\Test.txt","P")    'strMyResult = C:\Folder\
strMyResult = NameSplit("C:\Folder\Test.txt","N")    'strMyResult = Test
strMyResult = NameSplit("C:\Folder\Test.txt","E")    'strMyResult = txt
strMyResult = NameSplit("C:\Folder\Test","E")       'strMyResult = ""
```

Related Functions

Command: TTD | Function: Char | Function: FCNo | Function: Idx | Function: Len | Function: PU | Function: Str | Function: Textaddition (+) | Function: TTR (Text to Real) | Function: Val

Function: Idx

Function: Idx

Determines the index of a text in a DIAdem vector variable.

```
ReturnValue = Idx(String, VariableName [:Start])
```

Input Parameters

String	Specifies the text to search for.
VariableName	Specifies the name of the DIAdem vector variable to search.
[***Start***]	Specifies the index where DIAdem starts searching in the vector variable.

Return Parameters

ReturnValue	Returns the index of the first vector element that contains the text you specify, as a numeric value. The `Idx` function first searches for vector elements identical to the text you specify. If the function does not find any identical vector elements, it searches for the text at the beginning of the texts in the vector elements. The function returns the result `0` if DIAdem cannot find any text or text element identical to the text you specify.

Examples

The following example determines the index of a text in the DIAdem vector variable `TV`, which you can fill with text in the DIAdem Calculator.

VBScript	Python

Copy script

```
TV(1) = "DIAdem"
TV(2) = "is a"
TV(3) = "very"
TV(4) = "useful"
TV(5) = "software"
TV(6) = "use"
TV(7) = "it"
```

```
Dim intMyResult
intMyResult = Idx("DIAdem","TV")        'intMyResult = 1
intMyResult = Idx("software","TV:3")    'intMyResult = 5
intMyResult = Idx("use","TV")           'intMyResult = 6
intMyResult = Idx("ware","TV")          'intMyResult = 0
```

Note You cannot use the `Idx` function to search in enumeration variables and dynamic enumeration variables for enumeration terms and dynamic enumeration terms. Use the Ord function to search enumeration variables and dynamic enumeration variables.

Related Functions

Command: TTD | Function: Char | Function: FCNo | Function: FileNameSplit | Function: Len | Function: PU | Function: Str | Function: Textaddition (+) | Function: TTR (Text to Real) | Function: Val

Function: Len

Function: Len

Determines the text length.

```
ReturnValue = Len(String)
```

Input Parameters

String	Specifies a text or a text variable.

Return Parameters

ReturnValue	Returns the number of characters as an integer value. DIAdem counts blanks at the beginning, at the end, and within the text.

Examples

The following example determines the number of characters in a text.

VBScript	Python

Copy script

```
Call Calculate ("L1= Len(""DIAdem  "")")    'L1 = 8
```

Note DIAdem only uses the DIAdem function `Len` if you use the FormulaCalc command otherwise DIAdem uses the VBS function Len.

Related Functions

Command: TTD | Function: Char | Function: FCNo | Function: FileNameSplit | Function: Idx | Function: PU | Function: Str | Function: Textaddition (+) | Function: TTR (Text to Real) | Function: Val

Function: PU

Function: PU

Formats the contents of a text variable.

```
ReturnValue = PU(TextVarName1, TextVarName2)
```

Input Parameters

TextVarName1	Specifies a text variable that contains either the text to be formatted or the name of a DIAdem variable. Separate the individual values with the `@` character.
TextVarName2	Specifies the text variable with the format instructions. You can format texts flush left with the wildcard `L` and flush right with the wildcard `R`.

Return Parameters

Returns a formatted text.

Examples

The following example formats the contents of the text variable `strValue` according to the format instructions in the text variable `strFormat`.

VBScript	Python

Copy script

```
Dim strValue, strFormat, strMyResult
strValue = "123.654@789.345"
strFormat = "dddd.d ddd.ddd"
strMyResult = PU(strValue,strFormat)      'strMyResult = 123.7 789.345
```

The following example refers to three variables to be formatted, in the `strValue` variable.

VBScript	Python

Copy script

```
L1 = 1
Dim strValue, strFormat, strMyResult
strValue = "CN(L1)@ChD(1,L1)@CL(L1)"
strFormat = "LLLL ddd.d dddd"
strMyResult = PU(strValue,strFormat)    'Time 123.5 1024
```

Note If you transfer variable names to text variables, you must use DIAdem variables.

Note Use the Str function to convert a numeric value into a text or to format a text.

Related Functions

Command: TTD | Function: Char | Function: FCNo | Function: FileNameSplit | Function: Idx | Function: Len | Function: Str | Function: Textaddition (+) | Function: TTR (Text to Real) | Function: Val

Function: Str

Function: Str

Converts a numeric value into a text or formats a text.

```
ReturnValue = Str(String, [Format])
```

Input Parameters

String	Specifies a numeric value, a text, or a variable.
[***Format***]	Specifies the output format of the string.

String	Format	Example
Numeric value Numeric variable	Display format according to the format instructions	`d.dd` — Decimal with two places after the decimal point
Text Text variable	`L` for text flush to the left `R` for text flush to the right	`R:10` — Flush to the right and ten characters long `L:5` — Flush to the left

			and five characters long
	Time values Variant value with the subtype Date	Display format according to the format instructions for time values	`#MM/DD/YYYY hh:nn` — Date with hours and minutes specification.

Return Parameters

ReturnValue	Returns a formatted text. If you specify a format that has less places after the decimal point than the number to be converted, DIAdem rounds off the result.

Examples

The following example formats and converts a numeric value with a place after the decimal point.

VBScript	Python

Copy script

```
Dim sMyResult
sMyResult = Str(123.456,"d.d")  'sMyResult = "123.5"
```

The following example formats a text flush right and fills the rest of the specified places with blanks.

VBScript	Python

Copy script

```
Dim sMyResult
sMyResult = Str("DIAdem", "R:10")    'sMyResult = "    DIAdem"
```

The following example formats a text with leading zeros.

VBScript	Python

Copy script

```
Call MsgBoxDisp(Str(2.34,"0dd.dd")) ' => "02.34"
```

The first example determines the current calendar week and the second example formats the current date.

VBScript	Python

Copy script

```
Dim strCurrWeek, strMyDate
strCurrWeek = Str(TTR(CurrDate),"#k")
strMyDate   = Str((Now),"#YYYY, DD.T")
```

Note Some variables, for example CurrTime, CurrDate, and CurrDateTime, return time values as text. You can use the TTR function to convert these texts to numeric values.

Note Use the PU function to format the contents of a text or of a text variable.

Related Functions

Command: TTD | Function: Char | Function: FCNo | Function: FileNameSplit | Function: Idx | Function: Len | Function: PU | Function: Textaddition (+) | Function: TTR (Text to Real) | Function: Val

Calculator Functions > Text Functions > Function: Textaddition (+)

Function: Textaddition (+)

Links texts together to form a new text.

```
ReturnValue = String1 + String2
```

Input Parameters

String1	Specifies a text or the name of a text variable.
String2	Specifies a text or the name of a text variable.

Return Parameters

ReturnValue	Returns a text.

Examples

The following example links the contents of two text variables.

VBScript	Python

Copy script

```
T1 = "DIA"
```

```
T2 = "dem"
Call Calculate ("T1 = T1 + T2")    'T1 = DIAdem
```

Note Use the VBS link operator & to link texts in scripts.

Related Functions

Command: TTD | Function: Char | Function: FCNo | Function: FileNameSplit | Function: Idx | Function: Len | Function: PU | Function: Str | Function: TTR (Text to Real) | Function: Val

Function: TTR (Text to Real)

Function: TTR (Text to Real)

Converts a text in time format into a number in DIAdem double time format. The value depends on the variable ApplicationTimebaseHighResolution.

Use the command TTD instead of the `TTR` function in order to convert a text in time format into a VBS date.

Note If you save time values in DIAdem that are VBS VBDate subtype variables, the high resolution is lost. VBS only offers time values for this data type which provides only a limited resolution.

ReturnValue = `TTR(`***String*** `[,`***Format***`])`

Input Parameters

String	Specifies a text or a variable that contains the date in DIAdem time format. The text must not contain written or abbreviated weekdays, months, or calendar weeks. If the day or month is single-digit, it must contain leading zeros, for example, 01/02/2011.
[***Format***]	Specifies the time format for the text according to the format instructions. If you do not specify a time format, DIAdem applies the default format specified under **Settings»DIAdem Settings**.

Return Parameters

ReturnValue	Returns a Double value.

The RTT function converts a number that exists in the DIAdem double time format into a text.

Examples

The following example converts a date into a number.

VBScript	Python

Copy script

```
Dim dMyResult
dMyResult = TTR("09/11/1989","dd/mm/yyyy")
```

The following example determines the calendar week for the current date.

VBScript	Python

Copy script

```
Dim strCurrWeek
strCurrWeek = Str(TTR(CurrDate),"#k")
```

Related Functions

Command: ChnConvertNumericToTime | Command: ChnConvertTimeToNumeric | Command: ConvertTimeType | Command: TTD | Command: TTD | Function: Char | Function: FCNo | Function: FileNameSplit | Function: Idx | Function: Len | Function: PU |

Function: RTP (Real To Part of Date) | Function: RTT (Real to Text) | Function: Str | Function: Textaddition (+) | Function: Val

Function: Val

Function: Val

Determines a number in a text.

```
ReturnValue = Val(String, [ErrorValue])
```

Input Parameters

String	Specifies a text or a text variable.
[***ErrorValue***]	Specifies the numeric value that DIAdem returns if a conversion error occurs. If you do not specify `ErrorValue`, DIAdem displays an error message when a conversion error occurs.

Output Parameters

ReturnValue	Returns the first number, which is contained in the text, as a numeric value. If the text does not contain a number and if the second parameter is not set, DIAdem outputs an error message.

Examples

The following examples determines the first number in a text.

VBScript	Python

Copy script

```
Dim dblMyResult
```

```
dblMyResult = Val("3dimension")                'dblMyResult = 3
dblMyResult = Val("-3*2")                      'dblMyResult = -3
dblMyResult = Val("Points,5.12,7.03,11.7")     'dblMyResult = 5.12
dblMyresult = Val("OnlyText",0)                'dblMyresult = 0; No error message
```

Note Use the `Val` function instead of the VBS functions CDbl and CLng, to convert texts into numbers and to process these numbers in DIAdem. The VBS functions use the language set on your computer. The `Val` function always uses a point as the decimal character, which is customary in DIAdem.

Related Functions

Command: TTD | Function: Char | Function: FCNo | Function: FileNameSplit | Function: Idx | Function: Len | Function: PU | Function: Str | Function: Textaddition (+) | Function: TTR (Text to Real)

Calculator Functions > Command and Variable Functions

Command and Variable Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the DIAdem command and variable functions. You can use these functions to determine the type of a variable or of a command, or to read out the environment variables of the operating system.

Function: GetEnv

Function: GetEnv

Reads out an environment variable of the operating system.

```
ReturnValue = GetEnv(VariableName)
```

Input Parameters

VariableName	Specifies an environment variable.

Return Parameters

ReturnValue	Returns the content of the environment variable as text.

Examples

The following example reads out the `path` environment variable of the operating system.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = GetEnv("path")
```

Note Select the Windows menu **Start»Control Panel»System»Advanced»Environment Variables** to check which environment variables your Windows operating system uses.

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent | Command: VEnumIdx | Function: IdentifierType | Function: MaxOrd | Function: Ord | Function: ScriptEvaluate | Function: VEnum | Function: VIT | Function: VSA

Function: IdentifierType

Function: IdentifierType

Determines the type of a variable or a command.

```
ReturnValue = IdentifierType(CmdVarName)
```

Input Parameters

CmdVarName	Specifies a variable or a command.

Return Parameters

ReturnValue	Returns an integer value for the variable or command type:

Value	Meaning
0	Unknown type
1	DIAdem command registered by default
2	DIAdem command registered with GPI
3	User command
4	DIAdem variable registered by default
5	DIAdem variable registered with GPI
6	DIAdem user variable
7	Calculator function

Examples

The following example determines the type of the O1 variable.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = IdentifierType("O1")              'intMyResult = 4
```

The following example determines the type of the Calculate command.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = IdentifierType("Calculate")    'intMyResult = 1
```

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent | Command: VEnumIdx | Function: GetEnv | Function: MaxOrd | Function: Ord | Function: ScriptEvaluate | Function: VEnum | Function: VIT | Function: VSA

Function: MaxOrd

Function: MaxOrd

Determines the greatest index of enumeration and dynamic enumeration terms in enumeration variables or dynamic enumeration variables.

```
ReturnValue = MaxOrd(VariableName)
```

Input Parameters

VariableName	Specifies an enumeration variable or a dynamic enumeration variable. The property variables of the object-oriented script interface are constants and not enumeration variables or dynamic enumeration list variables.

Return Parameters

ReturnValue	Returns an integer for the index of the last enumeration or dynamic enumeration term. The first enumeration or dynamic enumeration term has the index `0`.

Examples

The following example determines the index of the last color in the ColorLst variable.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = MaxOrd("ColorLst")
```

The following example shows all the colors of the `ColorLst` variable in a message box.

VBScript	Python

Copy script

```
Dim intNo
For intNo = 1 To MaxOrd("Colorlst")
 Call MsgBoxDisp ("Color No " & intNo & " is " & VEnum("ColorLst",intNo))
Next
```

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent

| Command: VEnumIdx | Function: GetEnv | Function: IdentifierType | Function: Ord | Function: ScriptEvaluate | Function: VEnum | Function: VIT | Function: VSA

Function: Ord

Function: Ord

Determines the ordinal number in the ASCII character set for the first character of a text, and determines the index of the current enumeration or dynamic enumeration term for enumeration variables or dynamic enumeration variables.

```
ReturnValue = Ord(String [, Flag])
```

Input Parameters

<table>
<tr><td>String</td><td>Specifies a text, or an enumeration or a dynamic enumeration variable.</td></tr>
<tr><td rowspan="2">[Flag]</td><td>Specifies the evaluation of the string.</td></tr>
<tr><td>
<table>
<tr><th>Value</th><th>Meaning</th></tr>
<tr><td>0</td><td>DIAdem checks in the standard setting whether the string is a text or a variable.
If the string is a text, DIAdem evaluates the text.
If the string is a variable, DIAdem evaluates the current content of the variable.</td></tr>
<tr><td>1</td><td>DIAdem evaluates the string as a text.</td></tr>
</table>
</td></tr>
</table>

Return Parameters

ReturnValue	Returns an integer for the ASCII ordinal number of the first character in a string, or for the index of the current enumeration or dynamic enumeration term. The index of the first enumeration or dynamic enumeration term has the value `0`.

Examples

The following example determines the index of the term `"TIF"` in the variable ExType.

VBScript	Python

Copy script

```
Dim intMyValue
ExType = "TIF"
intMyValue = Ord("ExType")        'intMyValue = 3
```

The following example evaluates a variable as a text and returns the ASCII code of the first letter `E` .

VBScript	Python

Copy script

```
Dim intMyResult
ExType = "TIF"
intMyResult  = Ord("ExType",1)        'intMyResult  = 69
```

Note If you specify the variable name without quotation marks, DIAdem evaluates the current enumeration or dynamic enumeration term as a text. In this case the second example returns the value `84` as the result for the letter `T`.

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent | Command: VEnumIdx | Function: GetEnv | Function: IdentifierType | Function: MaxOrd | Function: ScriptEvaluate | Function: VEnum | Function: VIT | Function: VSA

Function: ScriptEvaluate

Function: ScriptEvaluate

Determines the contents of a user command in the DIAdem Calculator.

```
ReturnValue = ScriptEvaluate(UserCmdName)
```

Input Parameters

UserCmdName	Specifies a user command registered in DIAdem.

Return Parameters

ReturnValue	Returns the current contents of the user command as a text.

Examples (Calculator)

The following example assumes that you have registered the script in DIAdem with the user command `MyCommand`.

VBScript	Python

Copy script

```
Function MyCommand(MyPar)
  MyCommand = "Value: " & MyPar
End Function
```

The following example line determines the contents of the user command `MyCommand`.

VBScript	Python

Copy script

```
T1 = ScriptEvaluate("MyCommand(45.3)")     'T1 = Value: 45.3
```

Note You cannot use the `ScriptEvaluate` function in scripts.

Note Refer to Using User Commands for information on how to define your own commands.

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent | Command: VEnumIdx | Function: GetEnv | Function: IdentifierType | Function: MaxOrd | Function: Ord | Function: VEnum | Function: VIT | Function: VSA

Function: VEnum

Function: VEnum

Determines an enumeration or dynamic enumeration term in a script from a enumeration variable or dynamic enumeration variable.

```
ReturnValue = VEnum(VariableName, Index)
```

Input Parameters

VariableName	Specifies an enumeration variable or a dynamic enumeration variable. The property variables of the object-oriented script interface are constants and not enumeration variables or dynamic enumeration list variables.
Index	Specifies the index of an enumeration term or a dynamic enumeration term. Value range 0 to MaxOrd

Return Parameters

ReturnValue	Returns the enumeration term or dynamic enumeration term as text. If you enter an index that does not exist in the variable, DIAdem returns an empty text.

Examples

The following example determines the fifth color in the ColorLst variable.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = VEnum("ColorLst",4)    'strMyResult = "blue"
```

The following example shows all the colors of the `ColorLst` variable in a message box.

VBScript	Python

Copy script

```
Dim intNo
For intNo = 1 To MaxOrd("Colorlst")
  Call MsgBoxDisp ("Color No " & intNo & " is " & VEnum("ColorLst",intNo))
Next
```

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent

| Command: VEnumIdx | Function: GetEnv | Function: IdentifierType | Function: MaxOrd | Function: Ord | Function: ScriptEvaluate | Function: VIT | Function: VSA

Function: VIT

Function: VIT

Determines whether a user variable is enabled in DIAdem.

```
ReturnValue = VIT(VariableName)
```

Input Parameters

VariableName	Specifies a user variable.

Return Parameters

ReturnValue	Returns a Boolean value.

Examples

The following example determines whether the user variable `MyTest_` is enabled.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = VIT("MyTest_")
```

Note Use the commands GlobalDim and GlobalReDim to define global variables. You also can	

	define global variables in user command files. Declare the variable in the global range outside a function and register this user command file.

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent | Command: VEnumIdx | Function: GetEnv | Function: IdentifierType | Function: MaxOrd | Function: Ord | Function: ScriptEvaluate | Function: VEnum | Function: VSA

Function: VSA

Function: VSA

Determines whether a user variable file is enabled.

```
ReturnValue = VSA(UserVarFile)
```

Input Parameters

UserVarFile	Specifies a user variable file.

Return Parameters

ReturnValue	Returns a Boolean value.

	Note The user variable file must be located in the SCRIPT user path.

Examples

The following example determines whether the user variable file `MyVar.vas` is enabled.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = VSA("MyVar")
```

Note Use the commands GlobalDim and GlobalReDim to define global variables. You also can define global variables in user command files. Declare the variable in the global range outside a function and register this user command file.

Related Functions

Command: AddUserCommandToEvent | Command: RemoveUserCommandFromEvent | Command: VEnumIdx | Function: GetEnv | Function: IdentifierType | Function: MaxOrd | Function: Ord | Function: ScriptEvaluate | Function: VEnum | Function: VIT

Calculator Functions > Channel Functions

Channel Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the DIAdem channel functions. You can use these functions to determine characteristic channel values.

Function: CCh

Function: CCh

Calculates the characteristic values of a data channel.

```
ReturnValue = CCh(Channel, StatValueIndex)
```

Input Parameters

ChannelName	Specifies the data channel.

StatValueIndex	Specifies the index of the characteristic value to be calculated.

Index	Characteristic value
0	Mean value
1	Minimum
2	Maximum
3	Range
4	Total

Return Parameters

ReturnValue	Returns characteristic values of a data channel.

Examples

The following example determines the maximum value of the `Group1/Input` data channel.

VBScript	Python

Copy script

```
Dim dblMyResult
dblMyResult = CCh("Group1/Input",2)
```

Note Refer to Executing Channel Calculations in the Calculator and Assigning a Value to a Variable in the Calculator for more information about the Calculator.

Related Functions

Command: ChnFind | Command: ChnFindReverse | Function: FCNo | Function: Find | Function: FindReverse | Function: PNo

Function: FCNo

Function: FCNo

Determines the number of a channel from an external file header in a DAT file.

```
ReturnValue = FCNo(ChnName, [ChannelStart])
```

Input Parameters

ChnName	Specifies a channel name.
[***ChannelName***]	Specifies the channel number where DIAdem starts searching for the channel in the file header.

Return Parameters

ReturnValue	Returns the channel number of the channel you want. If DIAdem does not find a channel with the given name, the `FCNo` function delivers the value `0`.

Note To use the `FCNo` function, you must register the external file header. Use the HDLoad command to register external file headers in a script. Select **File»DAT Files»Import via Header** in **DIAdem NAVIGATOR** to open an overview of the external file headers in your computer. You can load, save, generate, and edit external file headers in this dialog box.

Examples

The following example uses the `HdLoad` command to load the `Auto1.dat` file header, which contains the data channels `Time, Deviation, Frequency,` and `Amplitude`. The `FCNo` function determines the channel number of the `Frequency` data channel.

VBScript	Python

Copy script

```
Dim intMyResult
Call HdLoad ("Auto1.dat")
intMyResult = FCNo("Frequency")     'intMyResult = 3
intMyResult = FCNo("Frequency",4)   'intMyResult = 0
```

Related Functions

Command: ChnFind | Command: ChnFindReverse | Function: CCh | Function: Find | Function: FindReverse | Function: PNo

Function: Find

Function: Find

Determines the first row in a data channel that meets a specified condition. The `Find` function searches a data channel from the beginning to the end.

```
ReturnValue = Find(Condition[,ChannelLine])
```

Input Parameters

Condition	Specifies the condition to check.
[***ChannelLine***]	Specifies the row number at which DIAdem begins checking the data channel.

Return Parameters

ReturnValue	Returns the first row number that satisfies the given condition. The `Find` function returns the result `0`, if no rows of the data channel satisfy the condition.

Examples

The following example searches the `Group1/Input` data channel of which the absolute value for the contents is higher than `10`. DIAdem starts checking from row 250 of the data channel and continues to the end of the channel.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = Find("Ch(""Group1/Input"")>10",250)
```

Note In the VBS syntax in the Calculator, use the ChnFind function instead of the `Find` function, to access the entire VBS functionality and user commands.

Related Functions

Command: ChnFind | Command: ChnFindReverse | Function: CCh | Function: FCNo | Function: FindReverse | Function: PNo

Examples

Searching for Outliers with the ChnEvent Function

Function: FindReverse

Function: FindReverse

Determines the first row in a data channel that meets a specified condition. The `ReverseFind` function searches a data channel from the end to the beginning.

```
ReturnValue = FindReverse(Condition[,ChannelLine])
```

Input Parameters

Condition	Specifies the condition to check.
[***ChannelLine***]	Specifies the row number at which DIAdem begins checking the data channel, back to the beginning.

Return Parameters

ReturnValue	Returns the first row number that satisfies the given condition. The function returns the result `0`, if no rows of the data channel satisfy the condition.

Examples

The following example searches the `Group1/Input` data channel of which the absolute value for the contents is higher than `10`. DIAdem checks from row 1000 up to the beginning of the data channel.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = FindReverse("Ch(""Group1/Input"")>10",1000)
```

Note In the VBS syntax in the Calculator, use the ChnFindReverse function instead of the `FindReverse` function, to access the entire VBS functionality and user commands.

Related Functions

Command: ChnFind | Command: ChnFindReverse | Function: CCh | Function: FCNo | Function: Find | Function: PNo

Function: PNo

Function: PNo

Determines the row in a numeric channel or a text channel, of which the contents are closest to a specified value or text.

```
ReturnValue = PNo(Channel, Value)
```

Input Parameters

ChannelName	Specifies the channel.
Value	Specifies the value or text to search for.

Return Parameters

ReturnValue	Returns the number of the first row with contents that are closest to the specified value or text.

Examples

The following example searches in the numeric channel `Group1/Input` for the row with contents closest to the value `77`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = PNo("Group1/Input",77)
```

The following example searches in the text channel `Group1/Input` for the row with contents that are closest to the text `test`.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = PNo("Group1/Input","test")
```

Related Functions

Command: ChnFind | Command: ChnFindReverse | Function: CCh | Function: FCNo | Function: Find | Function: FindReverse

Calculator Functions > Folder and File Functions

Folder and File Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the DIAdem folder and file functions. You can use these functions to determine whether folders and files exist and how big they are.

Function: DataFileSize

Function: DataFileSize

Determines the size in bytes of a DIAdem file that is a .tdm, .tdms, or .dat type and the associated binary files that contain the bulk data.

```
ReturnValue = DataFileSize(FileName)
```

Input Parameters

FileName	Specifies a filename with path and extension. In the filename and extension, you can use ? as a wildcard for a single character and * for several characters.

	If you do not specify a path, DIAdem searches for the file in the DIAdem folder.

Return Parameters

ReturnValue	Returns an integer value. If DIAdem cannot uniquely determine or find the file, the function returns the value `0`.

Examples

The following example determines the size of a TDM file and the associated binary file.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = DataFileSize(DataReadPath & "Drive.tdm")
intMyResult = DataFileSize(ProgramDrv & "Example\Data\Course.tdm")
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: FileAttrCheck | Function: FileDateGet | Function: FileSize | Function: FilEx | Function: FolderAttrCheck | Function: FolderDateGet | Function: FolderExist | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FileAttrCheck

Function: FileAttrCheck

Checks the attributes of a file.

```
ReturnValue = FileAttrCheck(FileName, Attribute)
```

Input Parameters

FileName	Specifies a file with path and extension. In the filename and extension, you can use `?` as a wildcard for a single character and `*` for several characters. If you do not specify a path, DIAdem searches for the file in the DIAdem folder.
Attributes	Specifies the attribute to be checked. **Value** — **Meaning** `"faReadOnly"` — The file is write-protected. `"faArchive"` — The file has been modified. `"faHidden"` — The file is hidden. `"faSystem"` — The file is a system file for the operating system. `"faOffline"` — The file has been moved from an archiving system to an offline memory.

Return Parameters

ReturnValue	Returns a Boolean value. If DIAdem does not find the given file, the function

	returns the result `FALSE`.

Examples

The following example checks whether the report file `Example.dat` in the library folder is write-protected.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = FileAttrCheck(LayoutLibrPath & "Example.tdr","faReadOnly")
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: DataFileSize | Function: FileDateGet | Function: FileSize | Function: FilEx | Function: FolderAttrCheck | Function: FolderDateGet | Function: FolderExist | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FileDateGet

Function: FileDateGet

Determines what time DIAdem created a file, or the last time DIAdem saved or opened the file.

```
ReturnValue = FileDateGet(FileName, TimeAttribute)
```

Input Parameters

FileName	Specifies a file with path and extension. In the filename and extension, you can use `?` as a wildcard for a single character and `*` for several characters. If you do not specify a path, DIAdem searches for the file in the DIAdem folder.
TimeAttribute	Specifies the time attribute you want: **Value** / **Meaning** fdCreate — Time of creation fdModify — Time of the last modification fdAccess — Time of the last access

Return Parameters

ReturnValue	Returns a text with the specified time. The time format depends on the language of the operating system. To select the time format for a language that does not correspond to the language of the operating system, first execute the ApplicationSetLocale command. The function returns an empty text if DIAdem does not find the given file.

Examples

The following example determines what time DIAdem created the file `Data?.tdm`.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = FileDateGet(DataWritePath & "Data?.tdm","fdCreate")
```

The following example determines what time DIAdem last saved the report file `Example.tdr`.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = FileDateGet(LayoutLibrPath & "Example.tdr","fdModify")
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: DataFileSize | Function: FileAttrCheck | Function: FileSize | Function: FilEx | Function: FolderAttrCheck | Function: FolderDateGet | Function: FolderExist | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FileSize

Function: FileSize

Determines the size of a file in bytes.

```
ReturnValue = FileSize(FileName)
```

Input Parameters

FileName	Specifies a file with path and extension. In the filename and extension, you can use `?` as a wildcard for a single character and `*` for several characters. If you do not specify a path, DIAdem searches for the file in the DIAdem folder.

Return Parameters

ReturnValue	Returns an integer value. If DIAdem cannot uniquely determine or find the file, the function returns the value `0`.

Examples

The following example determines the size of files.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = FileSize(DataLibrPath & "Example.tdm")
intMyResult = FileSize(LayoutLibrPath & "Example.tdr")
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function:

DataFileSize | Function: FileAttrCheck | Function: FileDateGet | Function: FilEx | Function: FolderAttrCheck | Function: FolderDateGet | Function: FolderExist | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FilEx

Function: FilEx

Valid names: FileExist, FilEx

Checks whether a file exists in a folder.

```
ReturnValue = FilEx(FileName)
```

Input Parameters

FileName	Specifies a file with path and extension. In the filename and extension, you can use `?` as a wildcard for a single character and `*` for several characters. If you do not specify a path, DIAdem searches for the file in the DIAdem folder.

Return Parameters

ReturnValue	Returns a Boolean value. TRUE — File exists in the specified folder. FALSE — File does not exist in the specified folder.

Examples

The following example checks whether the `Diadem.exe` file is in the DIAdem folder.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = FileExist("diadem.exe")    'blnMyResult = TRUE
```

The following example checks whether the library folder contains any DAT files.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = FilEx(DataLibrPath & "*.dat")    'blnMyResult = TRUE
```

The following example checks, whether any TDM files are in the NAVIGATOR-user folder. If the `FilEx` function returns the value `TRUE`, DIAdem opens the dialog box for loading data files. If the `FileEx` function returns the value `FALSE`, a message box displays an error message.

VBScript	Python

Copy script

```
If FilEx(DataReadPath & "*.tdm") Then
  Call FileNameGet("NAVIGATOR","FileRead","*.TDM")
  FileImportFilter = FileDlgFilter
  Call DataFileLoad(FileDlgFileName, FileImportFilter)
Else
  MsgBoxDisp("No TDM file found")
End If
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: DataFileSize | Function: FileAttrCheck | Function: FileDateGet | Function: FileSize | Function: FolderAttrCheck | Function: FolderDateGet | Function: FolderExist | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FolderAttrCheck

Function: FolderAttrCheck

Checks the attributes of a folder.

```
ReturnValue = FolderAttrCheck(FolderName, Attribute)
```

Input Parameters

<table>
<tr><td>FolderName</td><td>Specifies a folder with path.
If you do not specify a path, DIAdem searches for the folder in the DIAdem folder.</td></tr>
<tr><td rowspan="2">Attributes</td><td>Specifies the attribute to be checked.</td></tr>
<tr><td><table>
<tr><th>Value</th><th>Meaning</th></tr>
<tr><td>faReadOnly</td><td>The folder is write-protected.</td></tr>
<tr><td>faArchive</td><td>The contents of the folder have been modified.</td></tr>
<tr><td>faHidden</td><td>The folder is hidden.</td></tr>
<tr><td>faSystem</td><td>The folder is a system folder.</td></tr>
</table></td></tr>
</table>

Return Parameters

ReturnValue	Returns a Boolean value. If DIAdem does not find the given folder, the function returns the result `FALSE`.

Example

The following example checks whether the DIAdem library folder is write-protected.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = FolderAttrCheck("libr","faReadOnly")
```

The following example checks, whether the `C:\MyFolder` data folder is write-protected. If the `FolderAttrCheck` function returns the value `FALSE`, a message box appears with an error message. If the `FolderAttrCheck` function returns the value `FALSE`, DIAdem opens the dialog box for saving data files.

VBScript	Python

Copy script

```
If FolderAttrCheck("C:\MyFolder","faReadOnly") then
  MsgBoxDisp("Folder is readonly!")
Else
  Call FileNameGet("NAVIGATOR", "FileWrite")
  Call DataFileSave(FileDlgFileName)
End If
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: DataFileSize | Function: FileAttrCheck | Function: FileDateGet | Function: FileSize | Function: FilEx | Function: FolderDateGet | Function: FolderExist | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FolderDateGet

Function: FolderDateGet

Determines what time DIAdem created a folder, or the last time DIAdem modified or opened the folder.

```
ReturnValue = FolderDateGet(FolderName, TimeAttribute)
```

Input Parameters

FolderName	Specifies a folder with path.
TimeAttribute	Specifies the time attribute you want: **Value** — **Meaning** fdCreate — Time of creation fdModify — Time of the last modification fdAccess — Time of the last access

Return Parameters

ReturnValue	Returns a text with the time in the form `dd.mm.yyyy hh:nn:ss`.

Examples

The following example determines the last access date for the demo data folder.

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = FolderDateGet(DataReadPath,"fdModify")
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: DataFileSize | Function: FileAttrCheck | Function: FileDateGet | Function: FileSize | Function: FilEx | Function: FolderAttrCheck | Function: FolderExist | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FolderExist

Function: FolderExist

Checks whether a folder exists.

```
ReturnValue = FolderExist(FolderName)
```

Input Parameters

FolderName	Specifies a folder with path. If you do not specify a complete path, DIAdem searches for the folder in the DIAdem folder.

Return Parameters

ReturnValue	Returns a Boolean value. TRUE — Folder exists in the specified path. FALSE — Folder does not exist in the specified path.

Examples

The following example checks whether the `C:\MyFolder` data folder exists. If the `FolderExist` function returns the value `TRUE`, DIAdem opens the dialog box for loading data files. If the `FolderExist` function returns the value `FALSE`, a message box appears with an error message.

VBScript	Python

Copy script

```
If FolderExist("C:\MyFolder") then
  Call FileNameGet("NAVIGATOR","FileRead","*.TDM")
  FileImportFilter = FileDlgFilter
  Call DataFileLoad(FileDlgFileName,FileImportFilter,"Load")
Else
  MsgBoxDisp("Folder not found")
```

```
End If
```

The following example checks whether the library folder exists. DIAdem searches for the folder in the DIAdem folder because the specified path is not complete.

VBScript	Python

Copy script

```
Dim blnMyResult
blnMyResult = FolderExist("libr")    'blnMyResult = TRUE
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: FolderAttrClear | Command: FolderAttrSet | Command: FolderCopy | Command: FolderCreate | Command: FolderDelete | Command: FolderMove | Command: FolderRename | Command: FolderUnzip | Command: FolderZip | Command: PathAddTrailingBackslash | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: DataFileSize | Function: FileAttrCheck | Function: FileDateGet | Function: FileSize | Function: FilEx | Function: FolderAttrCheck | Function: FolderDateGet | Function: FolderSize | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Function: FolderSize

Function: FolderSize

Determines the size of a folder, subfolders included, in bytes.

```
ReturnValue = FolderSize(FolderName)
```

Input Parameters

FolderName	Specifies a folder with path. If you do not specify a path, DIAdem searches for the folder in the DIAdem folder.

Return Parameters

ReturnValue	Returns an integer value. If DIAdem cannot uniquely determine or find the folder, the function returns the value `0`.

Examples

The following example determines the amount of memory that the files in the demo folder of DIAdem take up.

VBScript	Python

Copy script

```
Dim intMyResult
intMyResult = FolderSize(DataReadPath)
```

Related Functions

Command: ChnValExpand | Command: DataFileCopy | Command: DataFileDelete | Command: DataFileLoad | Command: DataFileLoadRed | Command: DataFileLoadSel | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename |

Command: DataFileSave | Command: DataFileSaveSel | Command: DataFileSelDlg | Command: DataFileSelDlg | Command: FileDlgShow | Command: FileNameGet | Command: FileNameGet | Command: PathDlgShow | Commands and Variables for Working with Data | Commands and Variables for Working with Files and Folders | DIAdem Library Paths | DIAdem Path Variables | DIAdem User Paths | Function: DataFileSize | Function: FileAttrCheck | Function: FileDateGet | Function: FileSize | Function: FilEx | Function: FolderAttrCheck | Function: FolderDateGet | Function: FolderExist | Variable: FileExportFilter | Variable: FileFilter | Variable: FileImportFilter

Calculator Functions > Text File Functions

Text File Functions

The subordinate topics contained in the tree on the contents tab of the Help describe the DIAdem text file functions. You can use these functions, for example, to open, to close, or to read out functions.

Function: FR

Function: FR

Reads a line from a text file.

```
ReturnValue = FR(FileName, LineNo)
```

Input Parameters

FileName	Specifies a text file with path and extension. If you do not specify a path, DIAdem searches for the text file in the DIAdem folder.
LineNo	Specifies the number of the line that DIAdem reads out.

Return Parameters

ReturnValue	Returns a text. If the given text line is a comment line, DIAdem returns an empty text. DIAdem discards any comments in a text line and returns the remainder of the text line as the result. The `FR` function only recognizes a comment by the characters `{` and `(*`. If the line number you enter for the `LineNo` parameter is higher than the number of lines in the text file, the `FR` function returns the value `EOF`.

Examples

The following examples read out lines from a fictitious ANSI text file `Test.txt`. The file is located in the DIAdem SCRIPT read path.

VBScript	Python

Copy script

```
***************
This is an
example
text {only for demonstration
```

VBScript	Python

Copy script

```
Dim strMyResult
strMyResult = FR(ScriptReadPath & "test.txt",1)     'strMyResult = ***************
```

```
strMyResult = FR(ScriptReadPath & "test.txt",4)     'strMyResult = text
strMyResult = FR(ScriptReadPath & "test.txt",7)     'strMyResult = EOF
```

Note The `FR` function does not support Unicode files. Use the TextFileReadLn function to read text files. These functions also support long text and Unicode characters. Use the TextFileWriteLn function to write to text files.

Related Functions

Command: DataFileDelete | Command: DataFileLstGet | Command: DataFileMove | Command: DataFileRename | Command: DirLstWrite | Command: FileAttrClear | Command: FileAttrSet | Command: FileClose | Command: FileCloseAll | Command: FileCopy | Command: FileDateSet | Command: FileDelete | Command: FileMove | Command: FileOpen | Command: FileRename | Command: TextFileEncoding | Command: TextFileEOL | Function: DataFileSize | Function: TextFileEOF | Function: TextFileError | Function: TextFileErrorTxt | Function: TextFileSeek

Function: TextFileClose

Function: TextFileClose

Closes a text file.

```
ReturnValue = TextFileClose(TextFileHandle)
```

Input Parameters

TextFileHandle	Specifies the handle of the text file. Use the return value of the TextFileOpen function. If you enter `-1`, DIAdem closes all text files opened with the `TextFileOpen` command. If you do not specify a path, DIAdem searches for the text file in the DIAdem folder.

Return Parameters

ReturnValue	Returns a numeric value. If an error occurs when the text file closes, the

	function returns the value –1.

Examples

The following example creates a text file in ANSI code, writes twenty lines in this file, and closes it.

VBScript	Python

Copy script

```
Dim LineNo, intMyHandle, intMyText, intMyError
intMyHandle = TextFileOpen("NewText.txt",eTextFileAttributeCreate OR
eTextFileAttributeWrite OR eTextFileAttributeANSI)
For LineNo = 1 To 20
  intMyText= TextfileWriteLn(intMyHandle, "Line " & LineNo)
Next
intMyError = TextFileClose(intMyHandle)
```

Related Functions

Command: ExecuteExclusiveBegin | Command: ExecuteExclusiveEnd | Command: ExecuteExclusiveEndAll | Command: TextFileEncoding | Command: TextFileEOL | Function: FR | Function: TextFileEOF | Function: TextFileError | Function: TextFileErrorTxt | Function: TextFileSeek

Function: TextFileEOF

Function: TextFileEOF

Checks whether DIAdem has reached the end of a file when reading a text file linewise.

```
ReturnValue = TextFileEOF(TextFileHandle)
```

Input Parameters

TextFileHandle	Specifies the handle of the text file. Use the return value of the TextFileOpen function.

Return Parameters

ReturnValue	Returns a Boolean value.

Examples

The following example opens the text file `NewText.txt` and reads the file linewise to the end. DIAdem displays each line of text in a message box.

VBScript	Python

Copy script

```
Dim intMyHandle, strMyText, intMyError
intMyHandle = TextFileOpen("NewText.txt", eTextFileAttributeRead)
Do While Not TextFileEOF(intMyHandle)
  strMyText = TextFileReadLn(intMyHandle)
  Call MsgBoxDisp(strMyText,"MB_OK")
Loop
intMyError = TextFileClose(intMyHandle)
```

Related Functions

Command: TextFileEncoding | Command: TextFileEOL | Function: FR | Function: TextFileError | Function: TextFileErrorTxt | Function: TextFileSeek

Function: TextFileError

Function: TextFileError

Checks whether an error occurs when you open, write, or read a text file.

```
ReturnValue = TextFileError(TextFileHandle)
```

Input Parameters

TextFileHandle	Specifies the handle of the text file. Use the return value of the TextFileOpen function.

Return Parameters

ReturnValue	Returns a numeric value. If an error occurs, the function returns the value `-1`.

Examples

The following example attempts to open the text file `OldText.txt`. However, DIAdem does not find this text file in the DIAdem folder and the TextFileOpen function returns an error. The example then jumps into the subroutine for dealing with errors and displays the error number and the associated error text.

VBScript	Python

Copy script

```
Dim intMyHandle, intMyError
intMyHandle = TextFileOpen("OldText.txt",eTextFileAttributeRead)
If intMyHandle = -1 Then
  Call MsgError(intMyHandle)
Else
  Call MsgBoxDisp( "Text file can be read.",_
                   "MB_OK","MsgTypeInformation",,5)
  intMyError = TextFileClose(intMyHandle)
```

```
End If

Sub MsgError(FileNo)
 Call MsgboxDisp("Error number: " & TextFileError(FileNo) & VbCrLf & _
 "error message: " & TextFileErrorTxT(FileNo)) 'Show error number and error message
End Sub
```

Related Functions

Command: TextFileEncoding | Command: TextFileEOL | Function: FR | Function: TextFileEOF | Function: TextFileErrorTxt | Function: TextFileSeek

Function: TextFileErrorTxt

Function: TextFileErrorTxt

Displays an error text if an error occurs when you open, write, or read a text file.

```
ReturnValue = TextFileErrorTxt(TextFileHandle)
```

Input Parameters

TextFileHandle	Specifies the handle of the text file. Use the return value of the TextFileOpen function.

Return Parameters

ReturnValue	Returns an error text.

Examples

The following example attempts to open the text file `OldText.txt`. However, DIAdem does not find this text file in the DIAdem folder and the TextFileOpen function returns an error. The example then jumps into the subroutine for dealing with errors and displays the error number and the associated error text.

VBScript	Python

Copy script

```
Dim intMyHandle, intMyError
intMyHandle = TextFileOpen("OldText.txt",eTextFileAttributeRead)
If intMyHandle = -1 Then
  Call MsgError(intMyHandle)
Else
  Call MsgBoxDisp( "Text file can be read.",_
                   "MB_OK","MsgTypeInformation",,5)
  intMyError = TextFileClose(intMyHandle)
End If

Sub MsgError(FileNo)
 Call MsgboxDisp("Error number: " & TextFileError(FileNo) & VbCrLf & _
 "error message: " & TextFileErrorTxT(FileNo)) 'Show error number and error message
End Sub
```

Related Functions

Command: TextFileEncoding | Command: TextFileEOL | Function: FR | Function: TextFileEOF | Function: TextFileError | Function: TextFileSeek

Command: TextFileEncoding

Display all Hide all

Command: TextFileEncoding

Specifies the character encoding of a file. The function uses the Byte Order Mark (BOM) of the file to specify the character encoding.

ReturnValue = `TextFileEncoding(FileName)`

Input Parameters

FileName	Specifies the filename.

Return Parameters

ReturnValue	The return value is an Integer variable type. Contains the character encoding

4	`eTextFileAttributeANSI`	Encoded in ANSI.
8	`eTextFileAttributeUnicode`	Encoded in Unicode.
16	`eTextFileAttributeUTF8`	Encoded in UTF8.

Examples

The following example specifies the character encoding of a file:

VBScript	Python

Copy script

```
Dim intMyEncode
intMyEncode = TextFileEncoding(ScriptReadPath & "NewText.txt")
Select Case intMyEncode
Case eTextFileAttributeANSI
  Call MsgBoxDisp("ANSI")
Case eTextFileAttributeUnicode
  Call MsgBoxDisp("Unicode")
Case eTextFileAttributeUTF8
  Call MsgBoxDisp("UTF8")
Case Else
  Call MsgBoxDisp("Error occured")
End Select
```

	Related Functions Command: TextFileEOL | Function: FR | Function: TextFileEOF | Function: TextFileError | Function: TextFileErrorTxt | Function: TextFileSeek

Command: TextFileEOL

Display all Hide all

Command: TextFileEOL

Specifies the end-of-line character of a text file. The function checks the end-of-line character of the first line. By transferring the end-of-line character to the TextFileOpen function, you can speed up the opening of a text file.

ReturnValue `= TextFileEOL(FileName,` ***TextFileAttribute***`)`

Input Parameters

FileName	Specifies the filename.
TextFileAttribute	Specifies how DIAdem creates or opens a text file. You can connect various attributes with logic operators.

Return Parameters

ReturnValue	Specifies the line end character. TextFileLineEnd type return value.

Examples

The following example specifies the end of line character of a file.

VBScript	Python

Copy script

```
Dim strFile, intLineCount, intMyHandle, strEOL, strMyText, intMyError
strFile = ScriptReadPath & "NewText.txt"
strEOL = TextFileEOL(strFile, eTextFileAttributeRead)
intMyHandle = TextFileOpen(strFile, eTextFileAttributeRead, strEOL)
intLineCount = 0
Do
  strMyText = TextFileReadLn(intMyHandle)
  If (Not IsNull(strMyText)) then
    intLineCount = intLineCount+1
    Call MsgBoxDisp(strMyText,"MB_OK")
  End If
Loop until (IsNull(strMyText))
intMyError = TextFileClose(intMyHandle)
```

Related Functions

Command: TextFileEncoding | Function: FR | Function: TextFileEOF | Function: TextFileError | Function: TextFileErrorTxt | Function: TextFileSeek

Command: TextFileOpen

Display all Hide all

Command: TextFileOpen

Opens or creates a text file.

ReturnValue `= TextFileOpen(FileName,` ***TextFileAttribute***`, [`***TextFileLineEnd***`])`

Input Parameters

FileName	Specifies the file name.
TextFileAttribute	Specifies how DIAdem creates or opens a text file. You can connect various attributes with logic operators.
[TextFileLineEnd]	Specifies the line end character.

Return Parameters

ReturnValue	Contains the file handle as a numeric value. If an error occurs when the text file is being opened or created, the function returns the value -1. The return value is a TextFileHandle type.

You can open a text file faster if you specify the character encoding with the TextFileEOL function and transfer it to the `TextFileOpen` function.

Examples

The following example creates a text file in ANSI code and writes twenty lines in this file. DIAdem saves the file in the DIAdem script folder.

VBScript	Python

Copy script

```
Dim LineNo, intMyHandle, intMyText, intMyError
intMyHandle = TextFileOpen(ScriptReadPath & "NewText.txt", eTextFileAttributeCreate
OR eTextFileAttributeWrite OR eTextFileAttributeANSI)
For LineNo = 1 To 20
  intMyText= TextfileWriteLn(intMyHandle, "Line " & LineNo)
Next
intMyError = TextFileClose(intMyHandle)
```

The following example reads a text file in ANSI code.

VBScript	Python

Copy script

```
Dim strFile, intLineCount, intMyHandle, strEOL, strMyText, intMyError
```

```
strFile = ScriptReadPath & "NewText.txt"
strEOL = TextFileEOL(strFile)
intMyHandle = TextFileOpen(strFile, eTextFileAttributeRead, strEOL)
intLineCount = 0
Do
  strMyText = TextFileReadLn(intMyHandle)
  If (Not IsNull(strMyText)) then
    intLineCount = intLineCount+1
    Call MsgBoxDisp(strMyText,"MB_OK")
  End If
Loop until (IsNull(strMyText))
intMyError = TextFileClose(intMyHandle)
```

Note If you open a file with the filename extension `.Lst`, DIAdem opens this file for the Serial evaluation. DIAdem automatically accesses the files that have the same name as the script file, but have the filename extension `.Lst`. Therefore, you must open this file with the `TfDenyNone` attribute.

Related Topics

Command: ExecuteExclusiveBegin | Command: ExecuteExclusiveEnd | Command: ExecuteExclusiveEndAll

Function: TextFileReadLn

Function: TextFileReadLn

Reads a text file line up to the end of line identifier. DIAdem starts to read from the first line. After DIAdem reads the line, DIAdem positions the bookmark in the next line. The next time you call the `TextFileReadLn` function, DIAdem automatically reads the next row. If you want to jump to a specific row in a text file, use the TextFileSeek function.

ReturnValue = `TextFileReadLn(`***TextFileHandle***`)`

Input Parameters

TextFileHandle	Specifies the handle of the text file.

	Use the return value of the TextFileOpen function.

Return Parameters

ReturnValue	Returns the read text line without the line end identifier. When DIAdem reaches the file end, the function returns NULL.

Examples

The following example opens the text file `NewText.txt` and reads the file linewise to the end. The function checks for the value NULL to detect the file end.

VBScript	Python

Copy script

```
Dim strFile, intLineCount, intMyHandle, strEOL, strMyText, intMyError
strFile = ScriptReadPath & "NewText.txt"
strEOL = TextFileEOL(strFile)
intMyHandle = TextFileOpen(strFile, eTextFileAttributeRead, strEOL)
intLineCount = 0
Do
  strMyText = TextFileReadLn(intMyHandle)
  If (Not IsNull(strMyText)) then
    intLineCount = intLineCount+1
    Call MsgBoxDisp(strMyText,"MB_OK")
  End If
Loop until (IsNull(strMyText))
intMyError = TextFileClose(intMyHandle)
```

Related Functions

Command: ExecuteExclusiveBegin | Command: ExecuteExclusiveEnd | Command: ExecuteExclusiveEndAll | Command: TextFileEncoding | Command: TextFileEOL | Function: FR | Function: TextFileEOF | Function: TextFileError | Function: TextFileErrorTxt | Function: TextFileSeek

Function: TextFileSeek

Function: TextFileSeek

Jumps to a specific row in a text file.
Using `TextFileSeek` can lead to greater runtimes for large files. If you want to read a text file line by line, use the TextFileReadln function. After DIAdem reads the line, DIAdem positions the bookmark in the next line. The next time you call the `TextFileReadLn` function, DIAdem automatically reads the next row.

```
ReturnValue = TextFileSeek(TextFileHandle, LineNo)
```

Input Parameters

TextFileHandle	Specifies the handle of the text file. Use the return value of the TextFileOpen function.
LineNo	Specifies the line number.

Return Parameters

ReturnValue	Returns a numeric value. If an error occurs, the function returns either the value `-1` or a value that does is not the same as the specified line number.

Examples

The following example opens the text file `NewText.txt`, skips to the tenth row, and reads the file linewise to the end. DIAdem displays each line of text in a message box.

VBScript	Python

Copy script

```
Dim GotoLine, intMyHandle, intMyText, strMyText, intMyError
```

```
intMyHandle = TextFileOpen("NewText.txt",eTextFileAttributeRead)
GotoLine = 10
intMyError = TextFileSeek(intMyHandle, GotoLine)
If intMyError = Gotoline then
    strMyText = TextFileReadLn(intMyHandle)
    Call MsgBoxDisp(strMyText,"MB_OK")
End If
intMyError = TextFileClose(intMyHandle)
```

Related Functions

Command: TextFileEncoding | Command: TextFileEOL | Function: FR | Function: TextFileEOF | Function: TextFileError | Function: TextFileErrorTxt

Function: TextFileWriteLn

Function: TextFileWriteLn

Writes a text line into a text file.

```
ReturnValue = TextFileWriteLn(TextFileHandle, Text)
```

Input Parameters

TextFileHandle	Specifies the handle of the text file. Use the return value of the TextFileOpen function.
Text	Specifies a text that DIAdem inserts into the text file.

Return Parameters

ReturnValue	Returns a numeric value. If an error occurs when the text is written, the function returns the value `-1`.

Examples

The following example creates a text file in ANSI code and writes twenty lines in this file.

VBScript	Python

Copy script

```
Dim LineNo, intMyHandle, intMyText, intMyError
intMyHandle = TextFileOpen("NewText.txt",eTextFileAttributeCreate OR
eTextFileAttributeWrite OR eTextFileAttributeANSI)
For LineNo = 1 To 20
  intMyText= TextfileWriteLn(intMyHandle, "Line " & LineNo)
Next
intMyError = TextFileClose(intMyHandle)
```

Related Functions

Command: ExecuteExclusiveBegin | Command: ExecuteExclusiveEnd | Command: ExecuteExclusiveEndAll | Command: TextFileEncoding | Command: TextFileEOL | Function: FR | Function: TextFileEOF | Function: TextFileError | Function: TextFileErrorTxt | Function: TextFileSeek

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